

Benchmarking AI Agents for Kaggle Competitions

A Fair Three-Way Comparison with Paired Significance Testing

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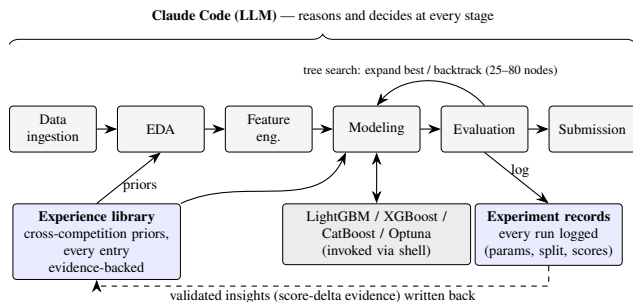
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The problem: agent benchmarks report noise

- ▶ LLM agents now solve Kaggle competitions end-to-end — but comparisons are “one run, one score, higher wins”
- ▶ Two questions current practice cannot answer:
 - **Positioning** — where does our agent stand *without* circular self-assessment?
 - **Noise** — score gaps sit at the 4th–5th decimal; are they real or sampling error?
- ▶ This work: benchmark **three methodologically distinct agents** on **18 competitions**, with a significance test that separates signal from noise

Three agents, three knowledge sources

	my-agent (ours)	AIDE	NVIDIA
Knowledge	experience library	none (cold start)	top-voted public kernel
Search	tree search, 25–80 nodes	LLM tree search, 20 steps	no search — reproduce
Validation	split by task type	no split legality	inherits kernel's



my-agent = Claude Code Skill:
the **LLM reasons and decides**;
GBDT tools execute.

Insights enter the experience
library **only with score-delta
evidence**.

Reference agents are **frozen** —
never improve your yardstick
mid-study.

A benchmark that cannot silently leak

- ▶ **18 competitions:** 13 Kaggle Playground + 5 older/niche ones; every submission scored on the **real leaderboard** (late submission)
- ▶ Three agents share one machine $\Rightarrow$ mechanically enforced isolation:
 - data — manifest-validated isolated roots
 - compute — mutex, one agent at a time
 - instructions — no recipes, no cross-agent hints; fix environments, never hint
- ▶ Every number traceable to structured experiment records

Method: a paired test with a free error scale

- ▶ Kaggle scores each submission on **two disjoint test subsets** (public / private) — two estimates of the same model; their gap = sampling noise
- ▶ Drift is **shared**: s5e10 drifts +0.00025/ + 0.00026/ + 0.00024 — common shift 0.00025, mutual scatter **0.00002**. Unpaired thresholds inflate noise $>10\times$
- ▶ Paired construction on the **between-agent gap** d (common drift cancels):

$$\text{sign}(d_{\text{priv}}) = \text{sign}(d_{\text{pub}})$$

$$|d_{\text{priv}}| > |d_{\text{priv}} - d_{\text{pub}}|$$

order replicates on disjoint samples

gap exceeds its own movement

Results: who wins depends on the evaluation set

54 duels, 40 decidable

my-agent	59%
AIDE	46%
NVIDIA	45%
undecidable	26%

a naive tally would report that 26% as
conclusions

Ranking reverses across sets

Win rate	Playground (13)	Full (18)
my-agent	44% (last)	59% (first)
NVIDIA	58% (first)	45% (last)

niche competitions have no good public kernels
⇒ reproduction loses its precondition

“Which agent is stronger?” is incomplete — ask “on what distribution of competitions?”

Reading the runs: why agents win and fail

- ▶ **AIDE — diagnosis cannot veto the score.** A bug fix silently switched to shuffled KFold on time-series data: metric jumped 18.4 → 4.4. Controlled re-run: **4.5× optimism bias**. Nine steps spent tuning a leaked metric — my-agent hit the same trap and quarantined it as “DIAGNOSTIC”
- ▶ **NVIDIA — votes $\neq$ quality.** An 89-vote workhorse kernel beat everyone; a 7,859-vote tutorial came last
- ▶ **Local CV $\neq$ leaderboard:** conclusions disagree in **10 of 13** competitions (s3e19: CV says we win big, leaderboard says we lose)

Takeaways

1. **Paired significance testing** for agent benchmarks — error scale comes free from Kaggle's public/private split; 26% of “verdicts” are noise without it
2. **CV-only benchmarks are unreliable** — submit to the real leaderboard (10/13 disagree)
3. **State the evaluation-set composition** — rankings are properties of method $\times$ competition distribution, not of the method alone

Ongoing: AIDE between-run variance; NLP/time-series/imaging extension; validation gates.

Code & experiment logs: github.com/930731haohao-afk/kaggle-agent